



Vanderbilt Center for Health Economic Modeling (VCHEM)

Est. 2020

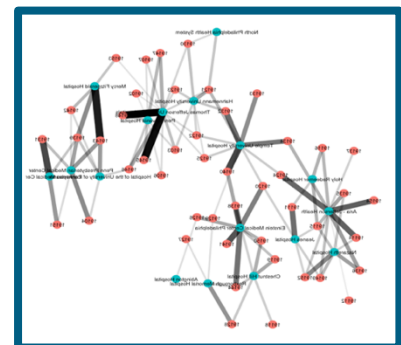
Our Mission is to Understand the Health and Economic Consequences of Health Policy and Technology Changes Worldwide Through Modeling By Harnessing the Diverse, Internationally Recognized Expertise of Vanderbilt Researchers.



Goal 1: To facilitate ongoing research in applied and theoretic modeling methods through internal and external institution collaborations.

The Center hosts the popular [web-based methods and code repository for using network analysis methods to define health care markets in the United States](#).

- This resource has been widely used by U.S. Government staff (Congressional Budget Office, Federal Trade Commission), researchers, and nonprofit organizations to guide policymaking and research on health care markets.
- Recent scholarship drawing from this Center-funded work includes new published articles on geographic health service areas and interfacility transfer networks for STEMI patients.¹



The center has facilitated several research partnerships at VUMC and elsewhere. Outside current collaborations include:



CDC Foundation
Together our impact is greater



Geisinger

¹ Nikpay, S., Leeberg, M., Kozhimannil, K., Ward, M., Wolfson, J., Graves, J., & Virnig, B. A. (2024). A proposed method for identifying Interfacility transfers in Medicare claims data. Health services research; Crook, H., Horta, M., Michelson, K. A., & Graves, J. A. (2024). Performance of health care service area definitions for capturing variation in inpatient care and social determinants of health. Health Services Research.

Center faculty and staff continue to showcase proficiency in modeling cost-effectiveness.



The Center's methodological advances initiated cross-institutional collaborations that led to recently funded R01 awards on pediatric service areas / markets and STEMI transfer networks.² We also expect a D43 training grant proposal submission to Fogarty International Center in Summer 2025.



Continued esteemed work in expertise areas: genomic screening,³ approaches to improving adherence to substance use disorder treatments among pregnant people, and approaches to reducing cardiovascular disease burden and improve the quality of care for stroke patients.⁴



Dr. Graves continues to draw on this Center-funded work for his role in ongoing high-profile research on high health care prices in the U.S., which is now under revision for a top economics journal (Journal of Political Economy).



The Center is now in the very early stages of building out a new research collaboration at VUMC centered on modeling the economic and health consequences of allocation policy changes for heart, lung and kidney transplants.

Goal 2: To serve as an intellectual hub for discovery, learning and dissemination for innovative applications and methods in health economic evaluation.

Center-developed activities and content are now integrated into the **Vanderbilt University School of Medicine curriculum**.

- Didactic material and case studies from the Center's funded international workshops are now integrated in the redesigned 1st and 2nd year longitudinal Foundations of Physician Responsibility (FPR), and in the fourth-year immersion course on Health Economics and Policy.
- Our **integration of methods and concepts in health economics into the curriculum is unique among medical schools in the U.S.**, and has earned high praise in recent student evaluations. Dr. Graves and colleagues expect to submit abstracts for short and long presentations on the overhaul of the health systems-oriented SOM curriculum to the upcoming AMA MedEx conference in Chicago in September 2025.

"This course had everything going against it, standing between 4th years and graduation. Despite that, Dr. Graves did an awesome job making this week engaging and informative with relevant topics. The reworked content was excellent. Great teaching on cost effectiveness analysis, global comparisons, and ways the insurers try to beat the system."

-4th year Medical Student

² Measuring and Mapping National Pediatric Acute Care Outcomes (R01HD112321), Eunice Kennedy Shriver National Institute of Child Health & Human Development; Disruptions in ST-Elevation Myocardial Infarction Transfer Networks and Cardiovascular Health of Rural Americans (R01HL153179).

³ Guzauskas, Gregory F., et al. "Population genomic screening for three common hereditary conditions: a cost-effectiveness analysis." *Annals of Internal Medicine* 176.5 (2023): 585-595.

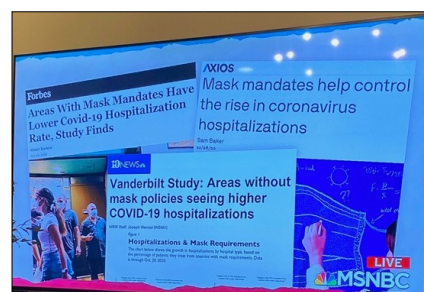
⁴ Zhu, J., Kamel, H., Gupta, A., Mushlin, A. I., Menzies, N. A., Gaziano, T. A., ... & Pandya, A. (2023). Prioritizing quality measures in acute stroke care: a cost-effectiveness analysis. *Annals of internal medicine*, 176(5), 649-657.

From 2020-2021, VCHEM was **utilized exclusively by the Governor's COVID-19 Unified Command** to provide analytic modeling support to the Department of Health and other state policymakers (legislature, governor's office, etc.).

- This work yielded **12 empirical policy briefs** that informed state and local policies on COVID-19, and that were frequently highlighted in **local**, state, national and international media.

VCHEM Impact During COVID-19

- Regular appearances and citations by local and regional leadership (Daily Metro briefings)
- **133k website visitors in 2020**, yielding more than 230k pageviews.
- Near daily interactions with local/state news outlets on state-specific data.
- **Considerable national and international media exposure** (see MSNBC screenshot).
- Publications in leading clinical journals (JAMA, JAMA Health Forum)

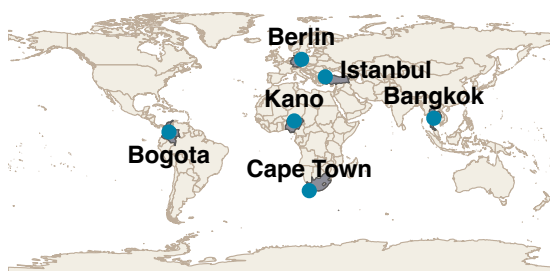


Goal 3: To further extend VU/VUMC's impact and inform the development and evaluation of technologic and policy changes worldwide

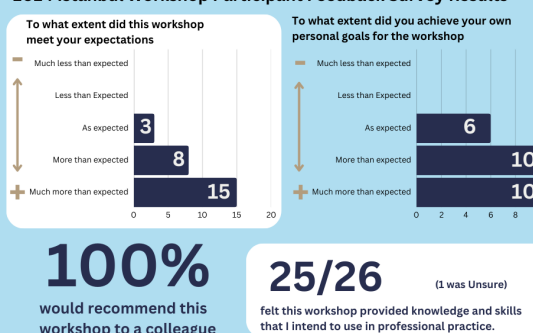
VCHEM faculty and staff have developed and led **six economic evaluation and modeling workshops** on nearly every continent.

- Workshop participants attended from dozens of countries worldwide, and included staff health economists at ministries of health and other government agencies, clinicians, academic researchers, and PhD students. Our curriculum is now fully developed, from beginner through intermediate and advanced methods.

Global Presence of VCHEM Workshops 2022–2024



2024 Istanbul Workshop Participant Feedback Survey Results



- In May 2025, and with new funding from Bloomberg Philanthropies, we will welcome **our first cohort of approximately 15 Fellows in Global Health Economics** to VUMC. These fellows will spend three weeks at Vanderbilt pursuing intensive mentorship, scholarship, and learning advanced health economic evaluation methods. This fellowship is co-administered by VUMC, Vital Strategies, and the CDC Foundation.